

ABHIJIT TADEPALLI

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SUMMARY

M.S. robotics researcher building real-time perception for manipulation: 3D tracking (Gaussian splatting), 6-DoF pose estimation (FoundationPose++), and automated hand-eye calibration on Franka Panda + Allegro/LEAP. Experienced in Python/C++, PyTorch, OpenCV, and modularizing legacy research code into hardware-validated pipelines.

EDUCATION

M.S. (Thesis), Robotics and Autonomous Systems

Arizona State University, Tempe, AZ

Expected May 2026

GPA: 3.63

Relevant coursework: Computer Vision, Modelling and Control of Robots, Robotic Systems II

B.Tech, Mechanical Engineering

Gandhi Institute of Technology and Management, Hyderabad, India

May 2024

Relevant coursework: Robotics, Artificial Intelligence and Machine Learning

EXPERIENCE

Graduate Researcher: IRIS Lab, Tempe, AZ

Dec 2024 – Present

- Extended and refactored a legacy Python/C++ 3D tracking pipeline: modular APIs, documentation, and new keyframe-selection strategies for 3D Gaussian-splatting tracking (ICRA 2026 submitted).
- Integrated FoundationPose++ for online 6-DoF pose estimation on real hardware (Franka Emika Panda, Allegro/LEAP) to support manipulation experiments.
- Built automated Python hand-eye calibration scripts, reducing experimental setup time by 70% and improving repeatability for multi-user trials.
- Owned lab-wide perception triage (tracking + pose): debugging failure cases and supporting multiple concurrent manipulation projects.

Machine Learning Intern: Bhabha Atomic Research Centre (BARC), Mumbai, India

May 2023 – Jul 2023

- Built an end-to-end PyTorch pipeline for steel-pipe defect inspection using YOLOv5, automating data capture/curation, training, and evaluation to accelerate NDT review by 80%.
- Tuned preprocessing for reflections and low-light conditions; achieved 90%+ validation accuracy on internal evaluation splits and improved detection consistency.

TECHNICAL SKILLS

Languages: C++, Python

Perception / ML: PyTorch, OpenCV

Dev Tools: Linux, Git

Robotics: ROS 1/2, MoveIt, Gazebo, MuJoCo, MATLAB/Simulink

Hardware: Franka Emika Panda, Allegro/LEAP Hand

Other: Unity, Blender, CAD, 3D printing, Arduino, CUDA (basic)

PUBLICATIONS

- TwinTrack: Bridging Vision and Contact Physics for Real-Time Tracking of Unknown Objects in Contact-Rich Scenes.** ICRA 2026 (submitted Sep 15, 2025). (*Contribution: 3D tracking pipeline refactor + keyframe selection; real-robot perception integration.*)

PROJECTS

Autonomous UAV Navigation and Vision-Based Landing: MATLAB, Simulink

Spring 2025

- Implemented a UAV guidance & control stack in MATLAB/Simulink: PID waypoint tracking with rigid-body dynamics; validated in Hardware-in-the-Loop (HIL).
- Designed a vision-based landing controller using color thresholding to detect a colored target platform and close the loop on landing; reduced trajectory deviation by 25% under simulated wind disturbances.

Room-Scale Spatial Mapping in VR: Meta Quest 3, Unity

Spring 2025

- Built a controller-driven room-scale mapping workflow in Unity (Meta Quest 3) to scan a room and reconstruct aligned boundaries/obstacles with geometry-respecting collision constraints.

ACTIVITIES

Graduate Instructional Aide & Course Grader: Arizona State University Tempe, AZ

Aug 2025 – Present

- Instructional aide and grader for *System Dynamics and Control II* and *Rigid Body Dynamics*; supported 400+ students via office hours, review sessions, and grading.